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About Me

I am a second-year Ph.D. student at the Department of Computer Science and Engineering (CSE) of Pennsylvania State University, advised by Prof. Dongwon Lee and co-advised by Prof. Kyusun Choi. Previously, I obtained my bachelor's and master's degrees from the Department of CSE, Bangladesh University of Engineering and Technology (BUET), where I also served as a Lecturer and Assistant Professor.

My research intersects **AI and cybersecurity**, with a particular focus on the **security and trustworthiness of Large Language Models (LLMs) and their surrounding ecosystem**. I am also co-leading one of ten globally selected red teams at the Amazon Nova AI Challenge 2026, working on autonomous multi-agent red teaming of web applications.

Research Interests

- **AI and Cybersecurity**: Security and trustworthiness of Large Language Models and their surrounding ecosystem.
- **Trustworthy and Reliable AI**: How knowledge and facts are stored, retrieved, and manipulated in Large Language Models; mechanistic interpretability, knowledge editing, and LLM unlearning.
- **Agentic AI for Red-Team Testing Automation**: Designing autonomous multi-agent systems for adversarial testing of AI-powered applications and web security.
- **LLM Security**: Overrefusal, jailbreaking, and robustness of LLM-based systems.
- **Bioinformatics and Health Informatics** (prior work): Deep learning for CRISPR off-target prediction and genomic analysis.

Education

- **Pennsylvania State University (PSU)** State College, PA, United States
  Ph.D. in Computer Science and Engineering; Passed Qualifying Exam August 2024 - Present
- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
  M.Sc. in Computer Science and Engineering; CGPA: 4.00/4.00 June 2019 - August 2023
- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
  B.Sc. in Computer Science and Engineering; CGPA: 3.97/4.00
  ∗ *Ranked **5th** in the class of 140 students.* February 2015 - April 2019

Publications

1. **Guardians of the Air: In-Device Detection of 5G Control-Plane Threats**
   Tianwei Wu, Abdullah Al Ishtiaq, Tianchang Yang, Yilu Dong, Kai Tu, Zeyu Song, Ridwanul Hasan Tanvir, **Md. Toufikuzzaman**, Shagufta Mehnaz, Syed Rafiul Hussain
   In *IEEE Symposium on Security and Privacy, 2026* [URL]
2. **From Control to Chaos: A Comprehensive Formal Analysis of 5G's Access Control**
   Mujtahid Akon, **Md. Toufikuzzaman**, Syed Rafiul Hussain
   In *IEEE Symposium on Security and Privacy, 2025* [URL] [Code]

3. **CRISPR-DIPOFF: An Interpretable Deep Learning Approach for CRISPR Cas-9 Off-Target Prediction**
   Md. Toufikuzzaman, Md. Abul Hassan Samee, M Sohel Rahman
   In *Briefings in Bioinformatics, 2024* [URL] [Code]

4. **A Deep Learning Based Semi-Supervised Network Intrusion Detection System Robust to Adversarial Attacks**
   Syed. Md. Mukit Rashid*, **Md. Toufikuzzaman***, Md. Shohrab Hossain
   {*Equal contribution}
   In *10th International Conference on Networking, Systems and Security (NSysS), 2023.* [URL][Code]

5. **Phylogenetic Analyses of SARS-CoV-2 Strains Reveal Its Link to the Spread of COVID-19 Across the Globe**
   Shashata Sawmya, Arpita Saha, Sadia Tasnim, Naser Anjum, **Md. Toufikuzzaman**, Ali Haisam Muhamad Rafid, Mohammad Saifur Rahman, M Sohel Rahman, Tanvir Alam
   In *MEDINFO 2021: One World, One Health–Global Partnership for Digital Innovation.* [URL] [Code]

6. **Fault Occurrence Detection and Classification of Fault Type in Electrical Power Transmission Line with Machine Learning Algorithms.**
   Shameem Hasan, **Md. Toufikuzzaman**
   In *International Journal on Electrical Engineering & Informatics 14, 3 (2022).* [URL]

7. **CRISPRpred (SEQ): a sequence-based method for sgRNA on target activity prediction using traditional machine learning**
   Ali Haisam Muhammad Rafid*, **Md. Toufikuzzaman***, Mohammad Saifur Rahman, M Sohel Rahman
   {*Equal contribution}
   In *BMC bioinformatics 21, 223 (2020).* [URL] [Code]

Research Experience (Selected)

1. **Precise LLM Unlearning via Bi-Level Optimization**: Designing a novel architecture and optimization framework for machine unlearning in Large Language Models. The key challenge is achieving precise, targeted forgetting of specific knowledge or facts while preserving model utility — without expensive full retraining. Our approach leverages bi-level optimization to jointly optimize the unlearning objective and the retain-set performance, and introduces a novel model architecture tailored for controlled knowledge removal.
   Supervisor: *Prof. Dongwon Lee and Prof. Ahmad Mousavi* Status: Under Review.

2. **Amazon Nova AI Challenge 2026 — Team Lion-0xA**: Co-leading Penn State's team as one of 10 global finalists. Our role as the Red Team involves designing autonomous multi-agent systems for web application security testing and user simulators for interacting with coding agents. Project title: *"Autonomous Co-Design and Multi-Agent Red Teaming for Advancing Trusted Agentic AI."*
   Faculty Adviser: *Prof. Dongwon Lee* Status: Ongoing (Final tournament: Sep. 2026).

3. **5G Access Control Formal Analysis**: Presented CoreScan, a formal analysis framework for verifying access control in 5G core networks. Uncovered five new classes of exploitable privilege escalation vulnerabilities using assume-guarantee reasoning and compositional verification. Findings were responsibly disclosed to GSMA.
   Supervisor: *Prof. Syed Rafiul Hussain* Status: Published in IEEE S&P 2025.

4. **CRISPR-DIPOFF**: Developed interpretable deep learning models (RNN and Transformer-based) for CRISPR Cas-9 off-target prediction, using evolutionary algorithms for hyperparameter tuning and integrated gradients for model interpretation. Outperformed state-of-the-art and extended established biological hypotheses on the sgRNA seed region.
   Supervisor: *Prof. M. Sohel Rahman* Status: Published in *Briefings in Bioinformatics, 2024.*

5. **NIDS**: Proposed a semi-supervised, adversarially robust deep learning approach for network intrusion detection, combining K-Means clustering, Autoencoders, Decision Trees, and MLPs. Validated robustness against FGSM and GAN-based IDSGAN attacks.
Supervisor: *Prof. Md. Shohrab Hossain* Status: Published in *NSysS, 2023.*

Employment

- **Pennsylvania State University (PSU)** State College, PA, United States
  Research Assistant, Department of CSE August 2024 – Present

Teaching Experience (Selected)

- **CMPEN 472:** Microprocessor and Embedded Systems *(Teaching Assistant)*
- Computer Security, Computer Networks, Computer Architecture, Database
- Information System Design, Software Development, Software Engineering, Simulation and Modeling
- Structured Programming, Object Oriented Programming, Data Structures and Algorithms, Digital Techniques

Honors & Awards

- Penn State College of Engineering Scholarship ($4,000), awarded as a competitive applicant for the Fall 2024 admission cycle.
- Dean's Award in each academic year for excellent undergraduate results: 2015 - 2019
- University Merit Scholarships in each semester for excellent undergraduate results: 2015 - 2019

Technical Skills

- **Programming Languages**: Python, C/C++, Java, Javascript, Rust
- **Database**: Oracle, SQL Server, MySQL, and Redis
- **Machine Learning Frameworks**: PyTorch, Huggingface, Keras
- **Vulnerability Assessment and Penetration Testing**: Metasploit, Burp-Suite, NMap, and different Web testing tools and frameworks.
- **Embedded Systems**: Arduino, ESP, ATMega32 and Different Micro-controllers and Networking chips
- **Development Frameworks**: Django, React

Selected Coursework

- Bioinformatics
- Machine Learning
- Pattern Recognition
- Data Mining
- Software Quality Assurance
- Distributed Computing Systems
- Computer Networks
- Network Security
- Computer Architecture
- Computer Vision
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Professional Services

- **Reviewer**
  IEEE Symposium on Security and Privacy (S&P), External Reviewer (2025)
  BMC Research Notes, Springer Nature (New York, US), (2020)

- **Coach**
  Coach of BUET teams for the International Collegiate Programming Contest (2020, 2021, 2022, 2023) and BUET Capture The Flag (CTF) teams of National Cyber Drills (2021, 2022).

References

Dongwon Lee	Kyusun Choi	M. Sohel Rahman
Professor	Professor	Professor
Department of IST, PSU	Department of CSE, PSU	Department of CSE, BUET
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